Automated Pet Feeder Project

Background

The objective of this project is to implement the Integrated Problem-Solving Process to design a programmable automated pet feeder intended for an animal shelter. This feeding mechanism should be able to dispense food for both cats and dogs at specified times, track their food consumption, and issue alerts in the event of problems, such as failure to dispense or the animals not eating. Although the final product will incorporate cost-effective hardware components like servo motors and sensors, the main emphasis of this task lies in the design and simulation of the system's logic. The assignment encompasses several key elements: assessing requirements, organizing information, creating an algorithm and flowchart, expressing the solution in clear terms, and testing it against various sample scenarios.

Step 1: Understand and define the problem

Through a thorough study of the case, the main objective is to design a low-cost, automated pet feeder system for a local animal shelter. To maintain consistent feeding routines for the animals, the feeding system must be able to

- Allocate food at scheduled periods of time, namely 8 a.m. and 6 p.m.
- Control food consumption by weighing the animal bowl before and after feeding animals.
- Examine whether the pet has eaten or not.
- Send alerts in case food has not been dispensed or provided in wrong amounts or remains uneaten after a certain period of time, which helps staff have a prompt response to feeding issues and keep the animal healthy.
- Use low-cost hardware

Essential inputs and outputs are described as follows:

Inputs	Usage
Feeding schedule	The user will be able to put specific times of each day, when the food be dispended
Food level scanner	Check if there’s enough food in the feeder through a sensor

Consumption sensor	Check the consumption of food through the other sensor in order to know whether the food has been eaten or is there any left
Manual feed button	In case there are some issues with the feeder, the user can always manually trigger the feeding system

Output	Usage
Motor control	Open/close the food dispenser
Alert system	Alert the user via light, sound or send direct message
Indicator	Show the food level status

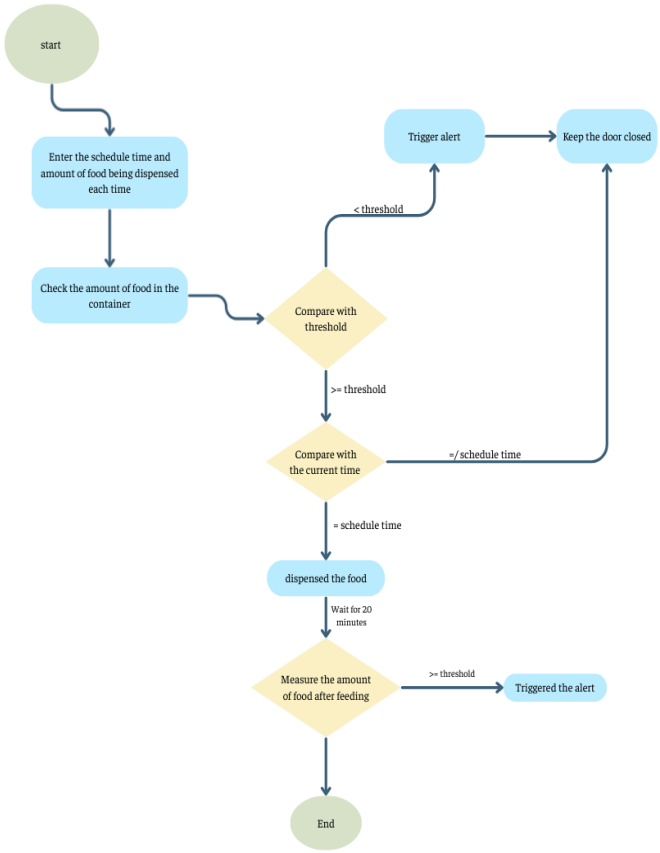
There are assumptions guiding the system design.

- One type of food will be provided for both cats and dogs.
- One pet will be served at each feeding station at a time.
- The feeding schedule will be fixed twice per day.
- The hardware will be limited to affordable components such as a servo motor for dispensing food, a weight sensor for monitoring consumption, and a real-time clock (RTC) module for time-based control.
- The power supply is stable, so it can run out of battery at any time.

Possible limitations are as follows:

- There is no support for wet food because of spoilage issues and the complexity of cleaning.
- Limited storage capacity (may need regular refilling)
- Cannot declare if food is spoiled.

A simple sketch of the flowchart:


start
Enter the schedule time and amount of food being dispensed each time
Check the amount of food in the container
Compare with threshold
< threshold
Trigger alert
Keep the door closed
>= threshold
Compare with the current time
=/ schedule time
= schedule time
dispensed the food
Wait for 20 minutes
Measure the amount of food after feeding
>= threshold
Triggered the alert
End